# Counterexample packet for arXiv:2501.12120

A surjective Banach-space affine isometry with no limit of $I^n(v)/n$

**Status.** **counterexample_likely_valid**. The construction below answers a specific uncertainty stated in Andres Navas, *Some examples of affine isometries of Banach spaces arising from 1-D dynamics*, arXiv:2501.12120. It does not answer the paper's broader Hilbert-space approximation Question 1 or group-realization Question 2.

**Source passage.** In Section 1, after the finite-dimensional discussion of affine isometries, the source notes that the vector limit $I^n(v)/n$ is known to exist for contractions of Hilbert spaces and for reflexive Banach spaces. It then gives a non-surjective isometric-map example on $\ell^1$ and says that there seems to be no example in the literature of a Banach-space isometry for which the limit does not exist, leaving the situation unclear.

**Idea of the proof.** The finite-dimensional argument is a mean-ergodic phenomenon. On $\ell^1(\mathbb{Z})$ the bilateral shift is an onto linear isometry, but its Cesaro averages on a basis vector are the normalized indicators of longer and longer intervals. Those normalized indicators have constant $\ell^1$ norm and drift through the space, so they are not Cauchy. Adding one basis vector as a translation turns this failure of mean convergence into the normalized orbit of a genuine surjective affine isometry.

**Theorem.** *There exists a surjective affine isometry $I$ of a Banach space and a vector $v$ such that the sequence $I^n(v)/n$ does not converge in norm.*

*Proof.* Let $X = \ell^1(\mathbb{Z})$ with its standard basis $(e_k)_{k\in\mathbb{Z}}$. Let $S : X \to X$ be the bilateral shift $Se_k = e_{k+1}$, and define

$$I(x) = Sx + e_0.$$

The map $S$ is a surjective linear isometry of $\ell^1(\mathbb{Z})$, hence $I$ is a surjective affine isometry.
For $n \geq 1$,

$$I^n(0) = e_0 + e_1 + \cdots + e_{n-1}.$$

Set

$$a_n = \frac{I^n(0)}{n} = \frac{1}{n}\sum_{j=0}^{n-1} e_j.$$

Then for every $n \geq 1$,

$$\begin{aligned}\|a_{2n} - a_n\|_1 &= \sum_{j=0}^{n-1}\left|\frac{1}{2n} - \frac{1}{n}\right| + \sum_{j=n}^{2n-1}\frac{1}{2n}\\ &= \frac{n}{2n} + \frac{n}{2n} = 1.\end{aligned}$$

Thus $(a_n)$ is not Cauchy, and so $I^n(0)/n$ does not converge in norm. $\square$

**Additional checks.** The example has metric drift 1 at 0, since

$$\frac{\|I^n(0)\|_1}{n} = \frac{\left\|\sum_{j=0}^{n-1} e_j\right\|_1}{n} = 1.$$

For any fixed $x \in \ell^1(\mathbb{Z})$,

$$\frac{I^n(x)}{n} = \frac{S^n x}{n} + \frac{I^n(0)}{n}, \qquad \left\| \frac{S^n x}{n} \right\|_1 = \frac{\|x\|_1}{n} \to 0,$$

so the same non-convergence holds for every starting point $x$.

**Novelty and literature search.** The run indexes had no entry for arXiv:2501.12120. The search phrases used locally and on the web were:

"$I^n(v)/n$" Banach space isometry limit does not exist,

affine isometry $\ell^1$ bilateral shift $I^n/n$,

linear rate of escape affine isometry Banach bilateral shift,

Kohlberg Neyman affine isometry $\ell^1$.

No exact prior occurrence of this explicit surjective affine-isometry example was found in that bounded search. Conceptually, the example is the standard failure of strong mean ergodic convergence for the bilateral shift on $\ell^1(\mathbb{Z})$, recast as an affine isometry.

**Human-review recommendation.** Review as a short counterexample to the source's unclear Banach-isometry case. The main point to verify is interpretive: the source's "isometry" means a surjective self-isometry, while the immediately preceding "isometric map" example is non-surjective.

# References


[1] A. Navas, *Some examples of affine isometries of Banach spaces arising from 1-D dynamics*, arXiv:2501.12120.

[2] E. Kohlberg and A. Neyman, *Asymptotic behavior of nonexpansive mappings in uniformly convex Banach spaces*, Amer. Math. Monthly 88 (1981), 698–700.